

B.Tech. Degree VI Semester Examination April 2018

ME 15 – 1606 (E3) ENERGY CONSERVATION AND ENVIRONMENT PROTECTION (2015 Scheme)

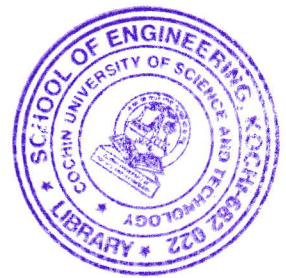
Time : 3 Hours

Maximum Marks : 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) List the strategies for better energy security of the nation.
- (b) List down the advantages of 'simple payback period' technique.
- (c) Give a typical energy audit reporting format.
- (d) Enumerate the methods of testing steam traps.
- (e) Name three factors affecting the boiler efficiency and explain briefly.
- (f) List the factors that affect energy efficiency in air compressors.
- (g) Write a short note on nuclear energy sources.
- (h) What is heat exchanger networking?
- (i) What are the implications of Global warming?
- (j) List the various steps to reduce particulates and water pollution.



PART B

(4 × 10 = 40)

- II. (a) Explain Internal Rate of Return (IRR) and Net Present Value method. (4)
- (b) How do an industry, nation and world would benefit from energy efficiency programs? (6)

OR

- III. (a) Annual savings after replacement of boiler for three years is ₹500000, ₹550000, ₹650000. Total project cost is ₹13.5 lakh. Considering cost of capital as 12%, what is the net present value of the proposal? (4)
- (b) Explain the trends in energy use pattern. (6)

- IV. Briefly list various energy conservation opportunities in a pumping and a refrigeration system. (10)

OR

- V. (a) How will you go about developing a waste heat recovery system? (5)
- (b) Distinguish between 'preliminary energy audit' and 'detailed energy audit' (5)

- VI. Write the cogeneration systems based on:

- (a) Diesel Engine. (5)
- (b) Gas Turbine. (5)

OR

- VII. (a) What is the importance of solar power in the present energy crisis in the world? (5)
- (b) Write a short note on wind energy. (5)

- VIII. (a) Describe the Greenhouse effect. (4)
- (b) Explain the types of air pollutants and their impacts. (6)

OR

- IX. (a) Briefly describe about the scope of environmental audit. (4)
- (b) Explain the detailed procedure of exhaust emission test in a vehicle. (6)